

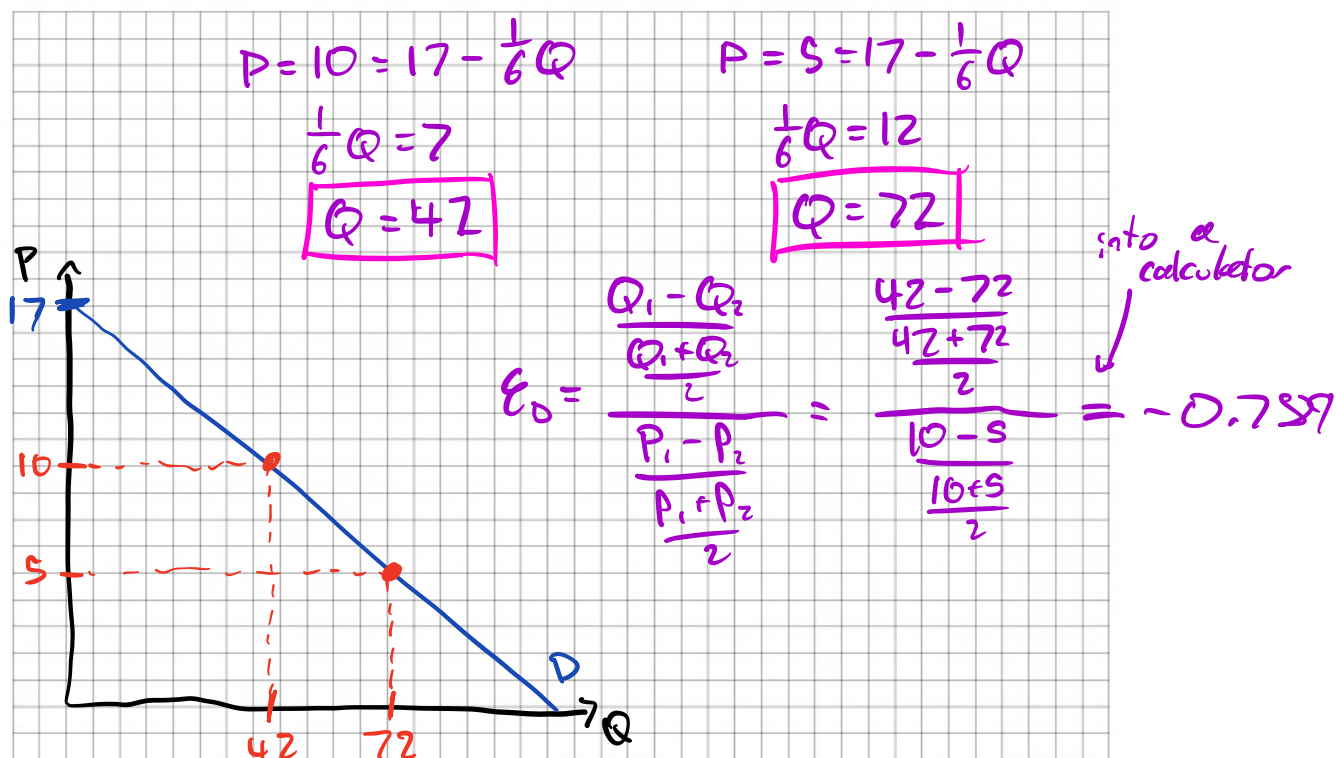
Honors Intro Micro | Homework 2

Due Tuesday Oct. 5

Homework is designed to both test your knowledge and challenge you to apply familiar concepts in new applications. Send me your questions at taylorjweidman@pitt.edu. Answer clearly and completely; show your work so I can understand your thought process for partial credit; you are welcomed and encouraged to work in groups as long as your work is your own.

Act 1

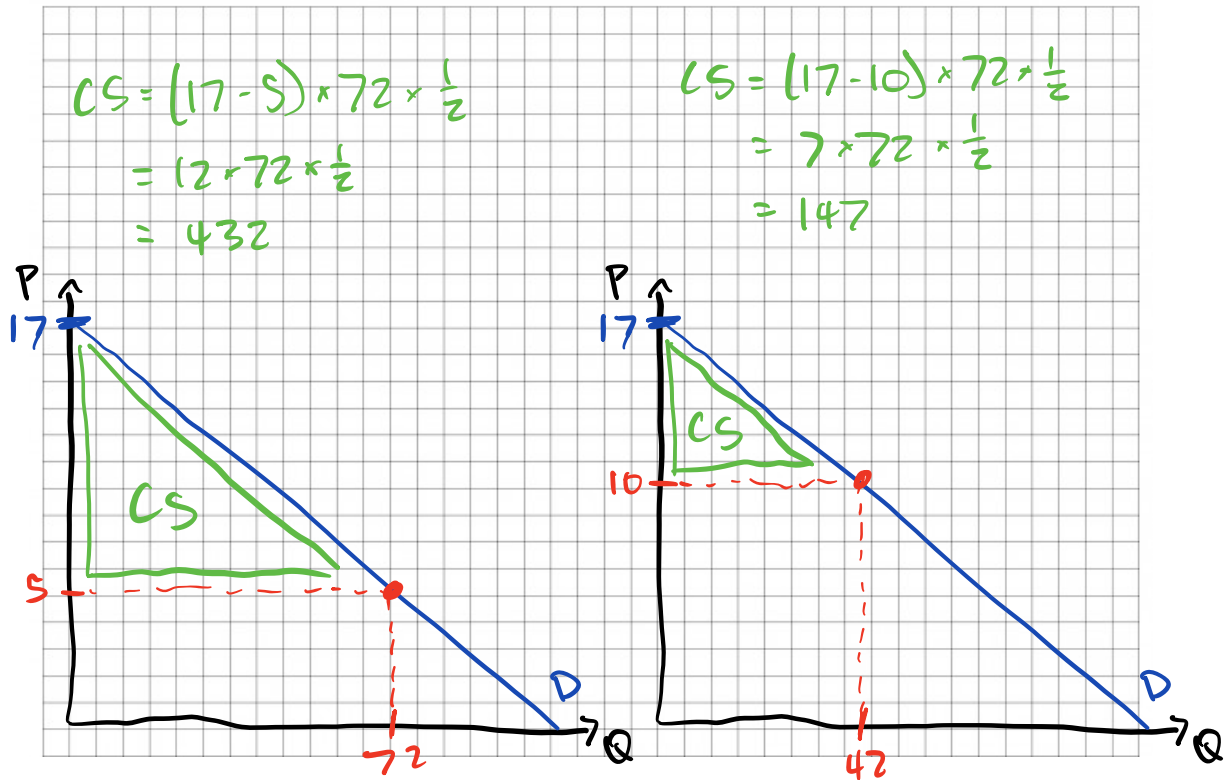
Wizards and witches tend to enjoy pumpkin pasties for the nostalgic taste. The demand curve for pumpkin pasties can be represented by $P = 17 - \frac{1}{6}Q$, in galleons. Use the midpoint method to find the elasticity of demand when the price changes from 5 to 10 galleons.



Elasticity of Demand: -0.789 This is inelastic.

Act 2

Use a graph to plot this demand curve, including the quantity demanded at both 5 galleons and 10 galleons. Then find and label the consumer surplus at these prices.



CS at 5 galleons: 432

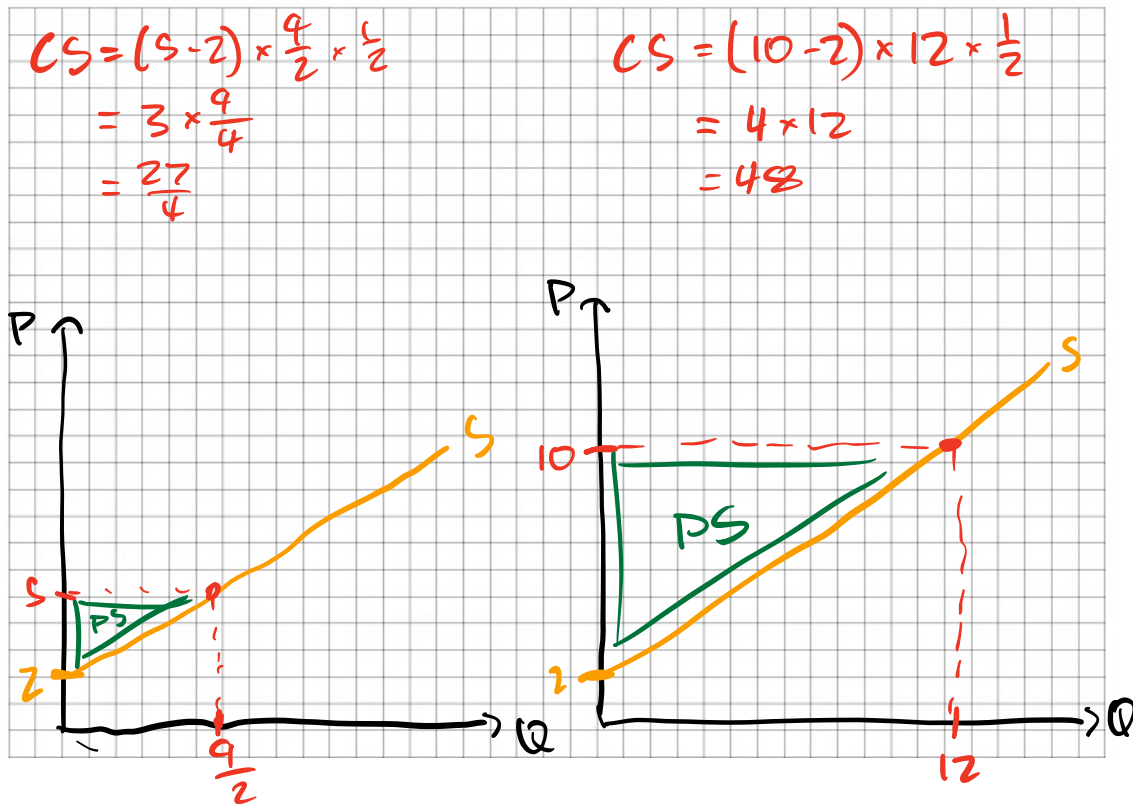
CS at 10 galleons: 147

Act 3

The supply curve for pumpkin pasties can be represented by the equation:

$$P = 2 + \frac{2}{3}Q$$

Use a graph to plot this supply curve, and find and label the producer surplus at both 5 galleons and 10 galleons.



PS at 5 galleons: $\frac{27}{4}$

PS at 10 galleons: 48

Act 4

The Ministry of Magic decided to institute a price ceiling of 10 galleons to allow many more to afford the treat. Use a graph of the demand curve and the supply curve to evaluate the welfare effects this policy had on the pumpkin pastie market.

Pre-policy

First, find equilibrium.

$$2 + \frac{2}{3}Q = 17 - \frac{1}{6}Q$$

$$\left(\frac{4}{6} + \frac{1}{6}\right)Q = 15$$

$$\frac{5}{6}Q = 15$$

$$Q = 18$$

$$P = 2 + \frac{2}{3} \times 18$$

$$P = 14$$

Second, find welfare.

$$CS = (17 - 14) \times 18 \times \frac{1}{2}$$

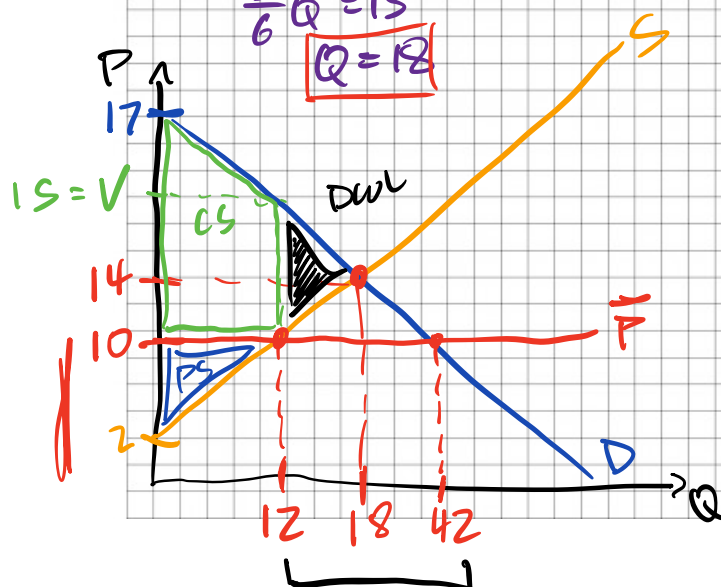
$$= 3 \times 9$$

$$CS = 27$$

$$PS = (14 - 2) \times 18 \times \frac{1}{2}$$

$$= 12 \times 9$$

$$PS = 108$$



Third, find the price ceiling.

$$10 = 2 + \frac{2}{3}Q_s$$

$$8 = \frac{2}{3}Q_s$$

$$12 = Q_s$$

$$10 = 17 - \frac{1}{6}Q_D$$

$$\frac{1}{6}Q_D = 7$$

$$Q_D = 42$$

Fourth, find the post-policy welfare. We need to find V.

$$P = 17 - \frac{1}{6} \cdot 12 = 15$$

$$CS = \Delta + \square$$

$$= (17 - 15) \times 12 \times \frac{1}{2} + (15 - 10) \times 12$$

$$= 2 \times 6 + 5 \times 12$$

$$CS = 72$$

$$PS = (10 - 2) \times 12 \times \frac{1}{2} = 8 \times 6$$

$$PS = 48$$

$$DWL = (15 - 10) \times (18 - 12) \times \frac{1}{2}$$

$$= 5 \times 6 \times \frac{1}{2}$$

$$DWL = 15$$

CS: 72

$\Delta CS: 72 - 27 = 45$

PS: 48

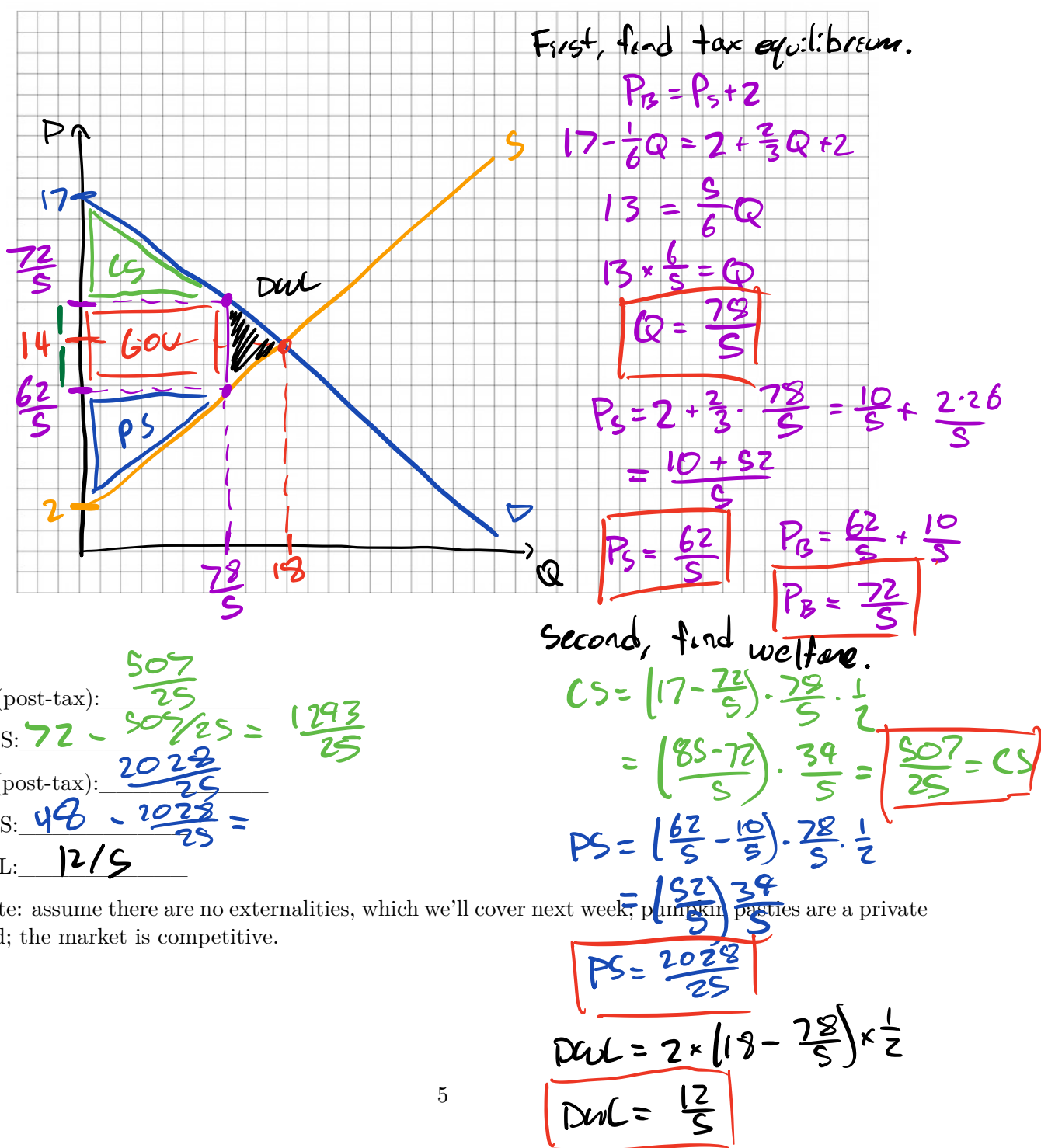
$\Delta PS: 48 - 108 = -60$

DWL: 15

Shortage/Excess: shortage of 30

Act 5

After years of careful epidemiological analysis, a subcommittee of the Ministry tasked with improving the health and wellbeing of the wizarding community published a story in the Daily Profit establishing a link between the consumption of pumpkin pasties and accidental magical spell casting by wizards and witches in public areas, with many cases of innocent muggles nearly being injured. To address these obvious public health concerns, the fiscal arm of the Ministry removed the price ceiling, instead imposing a 2 galleon tax on the sale of pumpkin pasties while reinvesting the revenues into researching a magical remedy for this diet-driven ailment.* Use a graph to illustrate the impact this tax had on the market.



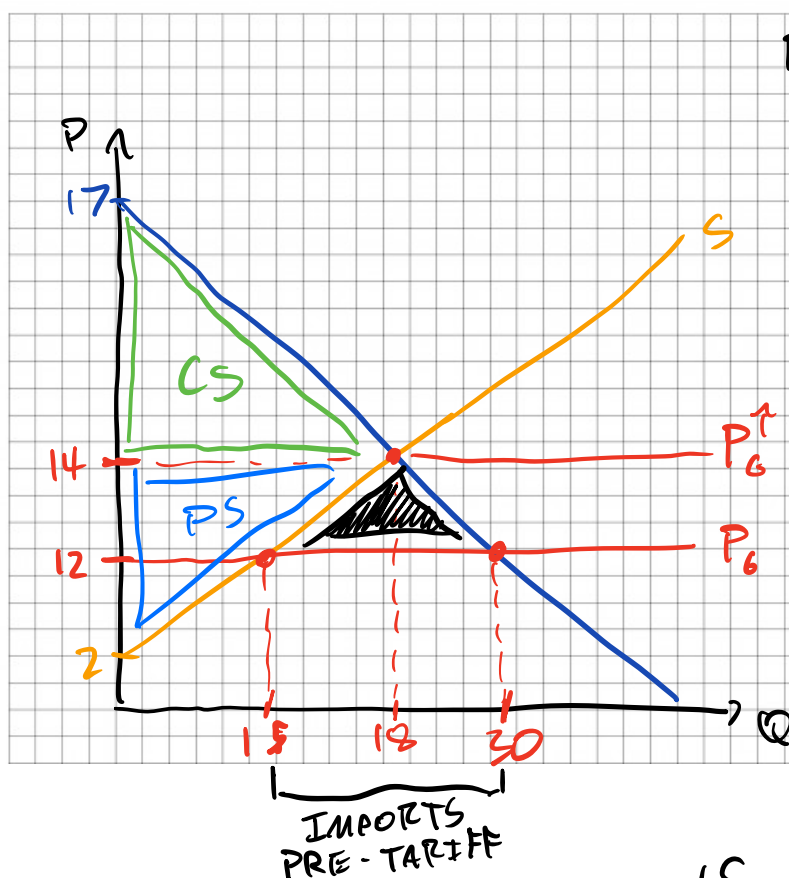
Act 6

Economic historians in the wizarding world remember a time before the Ministry had begun meddling with the domestic market. In those earlier days, trade barriers had been lifted for exports of pumpkin pasties to international magical markets with a world price of 12 galleons. At the time, the Ministry of Magic used this opportunity to pad its coffers without facing the political backlash associated with a domestic tax, instead imposing an import tariff of 2 galleons. Use a graph to illustrate what happened to the market after the tariff was imposed. Remember the supply and demand curves:

$$P = 17 - \frac{1}{6}Q$$

$$P = 2 + \frac{2}{3}Q$$

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First, we find domestic quantity supplied and quantity demanded.

$$12 = 2 + \frac{2}{3}Q_s$$

$$10 = \frac{2}{3}Q_s$$

$$Q_s = 15$$

$$12 = 17 - \frac{1}{6}Q_D$$

$$\frac{1}{6}Q_D = 5$$

$$Q_D = 30$$

Second, find DWL.

$$DWL = 2 \times 15 \times \frac{1}{2}$$

$$DWL = 15$$

What is the area of the deadweight loss from the tariff?

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